

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO1:** Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2  
Assessment Tool Weightage: 30%

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**QUESTIONS: 2**

**Total Marks:50**

Annexure A

Technical Presentation

| <b>Criteria</b>  | <b>Technical Content Accuracy(3)</b> | <b>Problem Identification(2.5)</b> | <b>Use of Tools(2)</b> | <b>Communication(1.5)</b> | <b>Professionalism(1)</b> | <b>Total(10)</b> |
|------------------|--------------------------------------|------------------------------------|------------------------|---------------------------|---------------------------|------------------|
| <b>Weightage</b> | <b>30%</b>                           | <b>25%</b>                         | <b>20%</b>             | <b>20%</b>                | <b>10%</b>                | <b>100%</b>      |
| <b>Q1</b>        |                                      |                                    |                        |                           |                           |                  |
| <b>Q2</b>        |                                      |                                    |                        |                           |                           |                  |

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### **Code of Conduct:**

I \_\_\_\_\_N.Vijetha (192311258)\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

N.Vijetha

## RUBRICS FOR CALCULATION:

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

## Annexure A

### Topic 1: Decision Support Systems (DSS)

#### Focus Areas:

- Definition and need of DSS
  - Components (Data, Model, UI)
  - Types of DSS
  - Real-world applications (banking, healthcare)
  - Advantages & limitations
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### Topic 2: Operational Systems vs DSS

#### Focus Areas:

1. OLTP vs OLAP concepts
  2. Key differences (data, users, purpose)
  3. Performance comparison
  4. Use-case scenarios
  5. Integration between systems
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### Topic 3: Architecture of Data Warehouse

#### Focus Areas:

1. Data warehouse definition
  2. Three-tier architecture
  3. ETL process
  4. Data marts & metadata
  5. Benefits in decision-making
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## Topic 4: OLAP and Data Cubes

### Focus Areas:

1. OLAP concept
  2. Data cube structure
  3. OLAP operations (slice, dice, roll-up, drill-down)
  4. Applications in analytics
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## Topic 5: Dimensional Data Modeling

### Focus Areas:

1. Fact and dimension tables
  2. Star schema vs snowflake schema
  3. Schema design process
  4. Real-world example
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## Topic 6: Query Processing & BI Tools

### Focus Areas:

1. Query processing stages
  2. Query optimization
  3. BI tools overview
  4. Reporting and visualization
  5. Examples of open-source BI tools
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## Topic 7: OLAP Application using KNIME

### Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation
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## PART B: PRESENTATION RULES & GUIDELINES

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### ◇ 1. Group Formation

1. Each group: 3–5 students
  2. Each member must participate
  3. Assign roles:
    1. Presenter
    2. Designer (PPT)
    3. Researcher
    4. Demo/Case handler
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### ◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
    1. Definition and importance
  2. Core Concepts
    1. Explanation with diagrams
  3. Process/Architecture
    1. Step-by-step explanation
  4. Real-World Example
    1. Case study or application
  5. Tool/Demo (if applicable)
    1. KNIME or BI tool
  6. Conclusion
    1. Summary + future scope
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### ◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
  1. Diagrams
  2. Flowcharts

3. Tables (for comparison)
  3. Avoid:
    1. Too much text
    2. Copy-paste content
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## ◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
  1. Eye contact
  2. Clear voice
  3. Logical flow



**SIMATS ENGINEERING**  
**SAVEETHA INSTITUTE OF MEDICAL AND**  
**TECHNICAL SCIENCES**



presentation

**ETL PROCESS**

**CSA1603– DATAWAREHOUSING AND DATAMINING**

**Guided By**  
Dr.D.Beulah David

**Presented by:**  
N.Vijetha(192311258)

# INTRODUCTION

## What is ETL?

ETL stands for:

- Extract
- Transform
- Load

It is the process of collecting data from different sources, cleaning and organizing it, and storing it in a Data Warehouse for analysis.

## Why ETL?

- Integrates data from multiple sources
- Improves data quality
- Supports Business Intelligence (BI)
- Enables faster decision-making



# STAGES OF THE ETL PROCESS

## Three Main Stages

### Extract

- Collects data from multiple sources.
- Retrieves structured and unstructured data.

### Transform

- Removes duplicate records.
- Corrects missing values.
- Standardizes formats.
- Validates data quality.

### Load

- Stores processed data in the Data Warehouse.
- Makes data available for reporting and analytics.

# ETL WORKFLOW



# REAL-TIME EXAMPLE

## ETL in an E-Commerce Company

### Extract

- ✓ Customer Orders
- ✓ Payment Details
- ✓ Product Information
- ✓ Delivery Records

### Outcome

- Sales Reports
- Customer Insights
- Product Recommendations

### Transform

- ✓ Remove duplicate orders
- ✓ Correct missing information
- ✓ Organize product categories

### Load

- ✓ Store cleaned data in the Data Warehouse.



# BENEFITS AND APPLICATIONS

## Benefits of ETL

- Improves data quality
- Integrates data from multiple sources
- Generates faster reports
- Supports accurate business decisions
- Saves time and reduces errors

## Applications

- Banking
- E-Commerce
- Healthcare
- Education
- Airline
- Social Media

# CONCLUSION

- ETL is a fundamental process in Data Warehousing.
- It collects data from multiple sources, cleans and transforms it into a consistent format, and stores it in a centralized Data Warehouse.
- ETL improves data quality, reduces errors, and ensures reliable information for analysis.
- It enables organizations to generate accurate reports, gain valuable insights, and make faster, data-driven decisions.



Engineer to Excel



*Thank You*